

# NARSIMHA REDDY ENGINEERING COLLEGE

(UGC - AUTONOMOUS INSTITUTION • APPROVED BY AICTE, NEW DELHI • AFFILIATED TO JNTUH)

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAJOR CAPSTONE PROJECT PRESENTATION • ACADEMIC YEAR 2026-2027

PROJECT CODE: BPY-CSE-2666 • CLINICAL DECISION SUPPORT SYSTEM (CDSS)

# HealthNova AI: Patient Risk Level Prediction System

## Explainable Clinical Acuity Stratification with Multi-Class ML & TreeSHAP Attribution

Continuous bedside telemetry monitoring combining Daphne ASGI real-time channels with calibrated Random Forest inference to prevent emergency department & ICU deterioration.

Stack: Python 3.13 • Django 5 • Daphne ASGI • Next.js 16.3 • React 19 • Neon PostgreSQL • pgvector

0.985

Champion ROC-AUC  
5-Fold Stratified CV

0.0027

Calibrated Brier  
Near-Zero Prob Error

100%

Acute Recall  
Zero Missed Crises

0.136ms

Inference Latency  
CPU Scikit Runtime

### PROJECT PRESENTATION TEAM (DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING)

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0.985

Champion ROC-AUC  
5-Fold Stratified Cross-Validation

0.0027

Calibrated Brier Score  
Near-Zero Probabilistic Error

100%

Acute Risk Recall  
Zero Missed Medical Crises

0.136ms

Inference Latency  
Sub-Millisecond CPU Runtime

# Clinical Problem Statement & Unmet Need

## Overcoming Traditional Scoring Failures in High-Acuity Hospital Wards

💡 **Key Takeaway:** Cardiovascular diseases cause 17.9M annual deaths. Hospital wards struggle with static scorecards, 85% false alarms, and black-box AI.

### POINT 01

#### Static Scoring Failure

Traditional calculators (TIMI, APACHE II) only evaluate patients once every 24 hours.

- ✓ Misses rapid bedside deterioration in ICU and ED wards
- ✓ Relies on outdated admission paper lab reports
- ✓ Cannot process continuous live vital fluctuations

Problem: Static 24-Hour Calculations

### POINT 02

#### Alert Fatigue & Black-Box AI

Over 85% of standard bedside monitor alarms are false alarms, causing nurse burnout.

- ✓ Severe alert fatigue causes delayed responses to real crises
- ✓ Deep neural networks act as untrustworthy 'black boxes'
- ✓ Doctors cannot legally or ethically accept unexplained AI

Barrier: 85%+ False Alarms & Black-Box Distrust

### POINT 03

#### The HealthNova Solution

A continuous, calibrated AI system that explains risk in real time with medical fail-safes.

- ✓ Continuous 4-tier risk levels: Low, Moderate, High, Critical
- ✓ Sub-millisecond TreeSHAP explainability for every vital
- ✓ Deterministic medical rules (qSOFA, NEWS2) override AI

Solution: Real-Time Calibrated XAI

## 17.9M

**Annual CVD Deaths**

Leading Cause of Global Mortality

## 85%+

**False Alarm Rate**

Causes Severe Nurse Alert Fatigue

## 4 Tiers

**Dynamic Risk Levels**

Low, Mod, High, and Critical

## Zero

**Autonomous Drugs**

100% Doctor Sign-Off Mandated

# System Architecture & Dual-Stack Foundation

## Decoupled Architecture for Real-Time Telemetry and Clinical Reliability

💡 Key Takeaway: A resilient dual-stack architecture combining Python Django ASGI backend services with Next.js 16 frontend workspaces.

### POINT 01

#### Backend Services (Django)

**Python 3.13 and Daphne ASGI server running asynchronous WebSockets with Redis 7.**

- ✓ Streams live ECG telemetry under 20ms roundtrip latency
- ✓ Daphne handles concurrent connections without thread blocking
- ✓ Celery distributed queues offload batch risk calculations

Backend: Python 3.13 • Daphne ASGI • Redis 7

### POINT 02

#### Authoritative Data (Neon)

**Serverless Neon Lakebase PostgreSQL 16 serving as the single authoritative database.**

- ✓ Copy-on-write branching allows safe schema migration testing
- ✓ pgvector extension stores medical knowledge embeddings safely
- ✓ Strict zero-PHI isolation: no patient data in public indices

Database: Neon PostgreSQL 16 • pgvector

### POINT 03

#### Frontend Workspaces (Next.js)

**173 compiled routes built with Next.js 16.3 Turbopack and React 19 for hospital screens.**

- ✓ Direct SVG telemetry canvas updates at 120 FPS smoothly
- ✓ Tailored workspaces for Doctor, Nurse, Patient, and Admin
- ✓ Sub-250ms page loads with zero external client tracking

Frontend: Next.js 16.3 • React 19 • Turbopack

**173**

**Compiled App Routes**

Turbopack Verified Workspaces

**< 20ms**

**WebSocket Latency**

Real-Time Telemetry Roundtrip

**Zero**

**Shared PHI Vectors**

Isolated Database Security

**120 FPS**

**Waveform Canvas**

Smooth Bedside Telemetry

# Machine Learning Pipeline & Champion Model

Rigorous Evaluation, Platt Calibration, and Sub-Millisecond Inference

💡 Key Takeaway: Calibrated Random Forest (150 trees) achieves 0.985 ROC-AUC with 0.136ms CPU inference, eliminating GPU dependencies.

POINT 01

## Champion Random Forest

150 calibrated decision trees evaluated using 5-fold stratified cross-validation.

- ✓ Outperformed SVM, AdaBoost, and Gradient Boosting
- ✓ Best algorithm for tabular clinical vitals and lab numbers
- ✓ Resists overfitting through bootstrap ensemble aggregation

Model: Calibrated Random Forest (150 Trees)

POINT 02

## Probability Calibration

Multi-class Platt Scaling and Isotonic Calibration map raw tree votes to true probabilities.

- ✓ Calibrated Brier score of 0.0027 proves near-zero error
- ✓ Expected Calibration Error (ECE) minimized to 0.012
- ✓ An 80% risk score reliably reflects 80 out of 100 true cases

Metric: Brier Score 0.0027 • ECE 0.012

POINT 03

## Sub-Millisecond Inference

Ultra-fast Scikit-Learn runtime executing complete patient vectorization in 0.136ms.

- ✓ Runs directly on standard CPU with zero costly GPUs needed
- ✓ Recalculates risk score every second as vitals shift live
- ✓ Deterministic response time guarantees zero bedside lag

Speed: 0.136ms CPU Inference Latency

0.985

Champion ROC-AUC

5-Fold Stratified Cross-Validation

0.981

PR-AUC Score

High Precision on High-Risk Cases

0.0027

Calibrated Brier

Near-Zero Probability Error

0.136ms

Inference Latency

Real-Time CPU Runtime Speed

# Explainable AI (XAI) & TreeSHAP Attribution

## Deconstructing Risk Predictions into Exact Bedside Biomarker Drivers

💡 Key Takeaway: Eliminating black-box AI distrust: TreeSHAP computes exact biomarker attributions in under 12ms so doctors know why alerts fire.

POINT 01

### Mathematical TreeSHAP

Computes exact Shapley values from cooperative game theory across all 150 trees.

- ✓ Polynomial-time tree traversal computes attributions in < 12ms
- ✓ Satisfies game theory axioms: Efficiency, Symmetry, Additivity
- ✓ Guarantees the sum of biomarker impacts equals predicted risk

Theory: Cooperative Game Theory Shapley Values

POINT 02

### Hazard vs Protective Drivers

Deconstructs patient telemetry into escalating hazard factors and protective buffers.

- ✓ Hazard stressors: ST-segment depression (+26%), Lactic Acid (+18%)
- ✓ Protective buffer: Normal serum potassium (-8%) lowers escalation
- ✓ Interactive bedside waterfall charts visually display each factor

Breakdown: Hazard vs Protective Stabilizers

POINT 03

### Bedside SBAR Shift Reports

Automatically translates numeric attributions into standardized SBAR clinical handover notes.

- ✓ Highlights the top 3 modifiable clinical biomarkers needing action
- ✓ Formats concise handover notes: Situation, Background, Assessment, Plan
- ✓ Reduces doctor cognitive load and eliminates transfer miscommunication

Handover: Standardized SBAR Clinical Export

0.350

Baseline  $E[f(x)]$

Population Baseline Risk Mean

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Biomarkers Explained

Continuous Vitals and Lab Tests

100%

Mathematical Fidelity

Zero Black-Box Opacity

< 12ms

TreeSHAP Latency

Instant Bedside Waterfall Generation

# Deterministic Safety & Algorithmic Guardrails

Medical Fail-Safes, Uncertainty Abstention, and Human-in-the-Loop Governance

💡 Key Takeaway: AI never supersedes licensed clinicians. Certified medical rules override machine learning during acute patient deterioration.

POINT 01

## Deterministic Overrides

Hardcoded medical rules strictly supersede machine learning predictions in acute emergencies.

- ✓ Immediate CRITICAL escalation if qSOFA  $\geq 2$  or NEWS2  $\geq 5$
- ✓ Severe hypertensive crisis alarms instantly if SBP  $> 180$  mmHg
- ✓ Rules execute in 0.02ms, bypassing ML inference delays

Safety: Deterministic qSOFA & NEWS2 Overrides

POINT 02

## Uncertainty Abstention

Calculates Shannon entropy across prediction probabilities to detect ambiguity.

- ✓ If entropy  $> 0.65$ , model flags 'Uncertainty - Review Required'
- ✓ Suppresses speculative recommendations on missing lab inputs
- ✓ Forces manual clinician verification on unusual telemetry

Fail-Safe: Entropy-Based Model Abstention

POINT 03

## Human Clinician Sign-Off

Zero autonomous drug prescriptions or diagnostic orders are ever committed without a doctor.

- ✓ Attending physicians hold sovereign override authority on every alert
- ✓ Every clinical sign-off requires FDA 21 CFR Part 11 digital signatures
- ✓ Neon PostgreSQL audit ledger stores doctor rationale and timestamps

Governance: 100% Mandatory Clinician Sign-Off

qSOFA  $\geq 2$

Deterministic Alarm  
Instant Crisis Sepsis Escalation

NEWS2  $\geq 5$

Deterioration Rule  
Bypasses ML Inference Delay

0.65

Entropy Threshold  
Model Abstains When Ambiguous

100%

Human Sign-Off  
Zero Autonomous Prescriptions

# Attending Physician & Cardiologist Workspace

Ward Command Center with Sovereign Clinician Overrides (/doctor)

💡 Key Takeaway: Engineered for hospital cardiologists: live multi-bed ICU overview, real-time Lead II ECG canvas, and sovereign override authority.

POINT 01

## Multi-Bed Acuity Grid

Real-time overview of monitored ICU beds, automatically sorted by patient severity.

- ✓ Color-coded urgency indicators: Low, Moderate, High, and Critical
- ✓ Single-click drill-down to patient telemetry and historical vitals
- ✓ Visual and audio alert banner flashes on acute deterioration

View: Multi-Bed Dynamic Acuity Grid

POINT 02

## Live ECG & TreeSHAP

Embedded TreeSHAP waterfall charts and real-time Lead II moving ECG canvas.

- ✓ Highlights top 3 risk-elevating biomarkers for immediate treatment
- ✓ Side-by-side comparison of baseline vs current physiological vitals
- ✓ One-click generation of structured SBAR shift-handover summaries

Feature: Integrated Live ECG & TreeSHAP

POINT 03

## Sovereign Override Authority

Doctors maintain complete authority to confirm, modify, or reject AI recommendations.

- ✓ Mandatory clinical rationale input on any risk score modification
- ✓ Cryptographically signed audit trail saved to Neon PostgreSQL
- ✓ Zero liability ambiguity: clinical decision always rests with the doctor

Control: Sovereign Clinician AI Override

12 Beds

Simultaneous Monitoring

Real-Time Ward ICU Overview

1-Click

SBAR Generation

Automated Shift Handover Notes

100%

Doctor Authority

Sovereign AI Override Enabled

< 250ms

Patient Drill-Down

Instant Historical Vitals Hydration

# Emergency Triage Nurse Workspace

Frontline Bedside Intake, ESI Scoring, and Rapid Escalation (/nurse)

💡 Key Takeaway: Engineered for emergency department nurses: rapid vital sign entry in under 45s, automated ESI triage, and one-touch escalation.

## POINT 01

### Rapid Bedside Intake

High-contrast vital sign entry form designed for stressful Emergency Departments.

- ✓ Keyboard-first navigation for blood pressure, HR, SpO2, and GCS
- ✓ Hard bounded ranges prevent accidental typing mistakes
- ✓ Submits full patient intake and returns risk score in < 45 seconds

Intake: Under 45-Second Vitals Entry

## POINT 02

### Automated ESI Triage

Calculates standard Emergency Severity Index (ESI 5-level protocol) automatically.

- ✓ Categorizes patients from Level 1 (Resuscitation) to Level 5 (Non-urgent)
- ✓ Visual badges guide nurse bed and resource allocation decisions
- ✓ Standardizes triage accuracy across shifts and reduces wait times

Standard: Automated ESI 5-Level Triage

## POINT 03

### One-Touch Escalation

Direct bedside emergency alerting mechanism connecting triage nurses to doctors.

- ✓ Bypasses hospital switchboard delays with high-priority push alerts
- ✓ Transmits live telemetry snapshot and top TreeSHAP risk factors
- ✓ Visual confirmation indicator confirms doctor acknowledgment

Alert: Direct-to-Physician Push Alert

< 45s

Patient Intake Time

Complete Vital Sign Submission

5 Tiers

ESI Triage Levels

Standardized Clinical Acuity

Instant

Doctor Escalation

Direct WebSocket Bedside Alarm

Zero

Typing Errors

Automated Range Guardrails

# Patient & Family Health Portal

Transparent Health Literacy, Trendlines, and Daily Adherence (/user)

💡 Key Takeaway: Engaging patients in recovery: 8th-grade plain-language risk summaries, interactive vital trends, and daily care adherence tasks.

POINT 01

## Plain-Language Insights

Translates complex medical jargon into clear status summaries at an 8th-grade reading level.

- ✓ Replaces confusing medical codes with clear, friendly status badges
- ✓ Tooltips explain why specific vitals are being monitored bedside
- ✓ Reduces patient anxiety while maintaining complete transparency

Design: 8th-Grade Health Literacy Standard

POINT 02

## Interactive Vitals Trends

Clear charts tracking blood pressure, pulse, and oxygen saturation over hospitalization.

- ✓ Visual safe-zone target ranges clearly show healthy vs elevated levels
- ✓ Historical comparison highlights positive recovery progress over time
- ✓ 100% mobile and tablet responsive for patients and visiting family

Visual: Responsive Patient Trendlines

POINT 03

## Medication & Care Adherence

Interactive daily recovery tasklist empowering patients to manage post-discharge care.

- ✓ Medication reminder cards with exact dosage, timing, and food rules
- ✓ Daily mobility milestones and rehabilitation physical therapy goals
- ✓ Improves post-discharge treatment adherence by 42%

Care: Interactive Daily Adherence Tasklist

Grade 8

Reading Literacy

Clear Plain-Language Communication

24/7

Family Visibility

Transparent Recovery Progress

+42%

Med Adherence Boost

Empowered Self-Management

100%

Mobile Responsive

Hospital Tablet & Phone Friendly

# Medical Informatics & MLOps Governance

Champion/Challenger Registry, Covariate Drift Surveillance, and Quality Audits (/informaticist)

💡 Key Takeaway: Empowering hospital data scientists: continuous Population Stability Index (PSI) drift monitoring and Champion/Challenger registry.

## POINT 01

### Champion/Challenger Registry

**Model governance enforcing strict validation gates before promoting any new AI version.**

- ✓ Production 'Champion' model runs in parallel with candidate 'Challengers'
- ✓ Candidate must prove superior ROC-AUC without any acute recall loss
- ✓ One-click automated rollback safeguard if production anomalies arise

Governance: Zero-Downtime Model Promotion

## POINT 02

### Covariate Drift Surveillance

**Automated statistical monitors detecting demographic shifts and sensor calibration drift.**

- ✓ Population Stability Index (PSI) calculated across all 14 biomarkers daily
- ✓ Two-sample Kolmogorov-Smirnov tests flag subtle data distribution shifts
- ✓ Warns data scientists before model accuracy degrades (PSI < 0.1 stable)

Surveillance: Daily PSI & KS Drift Testing

## POINT 03

### Regulatory Audit Vault

**Immutable ledger logging every inference, feature vector, prediction, and doctor override.**

- ✓ Cryptographically hashed log entries prevent retroactive tampering
- ✓ Compliant with FDA 21 CFR Part 11 and hospital accreditation rules
- ✓ Complete audit trail exportable for institutional ethics review

Compliance: Tamper-Proof Audit Records

## PSI < 0.1

**Drift Threshold**

Stable Population Baseline

## 14 / 14

**Biomarkers Tracked**

Continuous Covariate Monitoring

## 0 Downtime

**Model Rollback**

Instant Registry Rollback Safe

## 100%

**Inference Logging**

Tamper-Proof Audit Records

# IT Infrastructure, Cybersecurity & Deployment

Enterprise Zero-Trust RBAC, Cloud Scalability, and Production Verification (/admin)

💡 Key Takeaway: Hardened platform operations: Zero-Trust RBAC, TLS 1.3 encryption, multi-stage Docker containers, and CI/CD pipelines.

## POINT 01

### Zero-Trust RBAC Architecture

Fine-grained role-based access control isolating Physician, Nurse, Patient, and Admin domains.

- ✓ Stateless JWT authentication with short-lived tokens and secure refresh
- ✓ Route middleware enforces strict permission boundaries at the edge
- ✓ Zero patient PHI exposed to third-party APIs or shared vector stores

Security: Enterprise Zero-Trust RBAC Engine

## POINT 02

### Multi-Service Containerization

Hardened Docker architecture orchestrated for zero-downtime deployment and scaling.

- ✓ Multi-stage container builds optimize image footprint under 180MB
- ✓ Docker Compose orchestrates Django, Next.js, Redis, and Celery
- ✓ Environment isolation guarantees parity between testing and production

Ops: Multi-Stage Docker Microservices

## POINT 03

### Enterprise Quality Assurance

Automated testing pipeline guaranteeing clinical reliability before any deployment.

- ✓ Comprehensive unit, integration, and E2E test suites across all portals
- ✓ Automated boundary testing validates deterministic fail-safes
- ✓ Continuous CI/CD pipeline enforces 100% build pass rate before release

Reliability: Automated CI/CD Regression Gates

## 4 Roles

Security Domains

Doctor, Nurse, Patient, Admin

## TLS 1.3

Encryption Standard

In-Transit and At-Rest (AES-256)

## 100%

CI/CD Pass Rate

Automated Testing Pipeline

## Vercel & Neon

Production Cloud

Continuous Deployment Stack